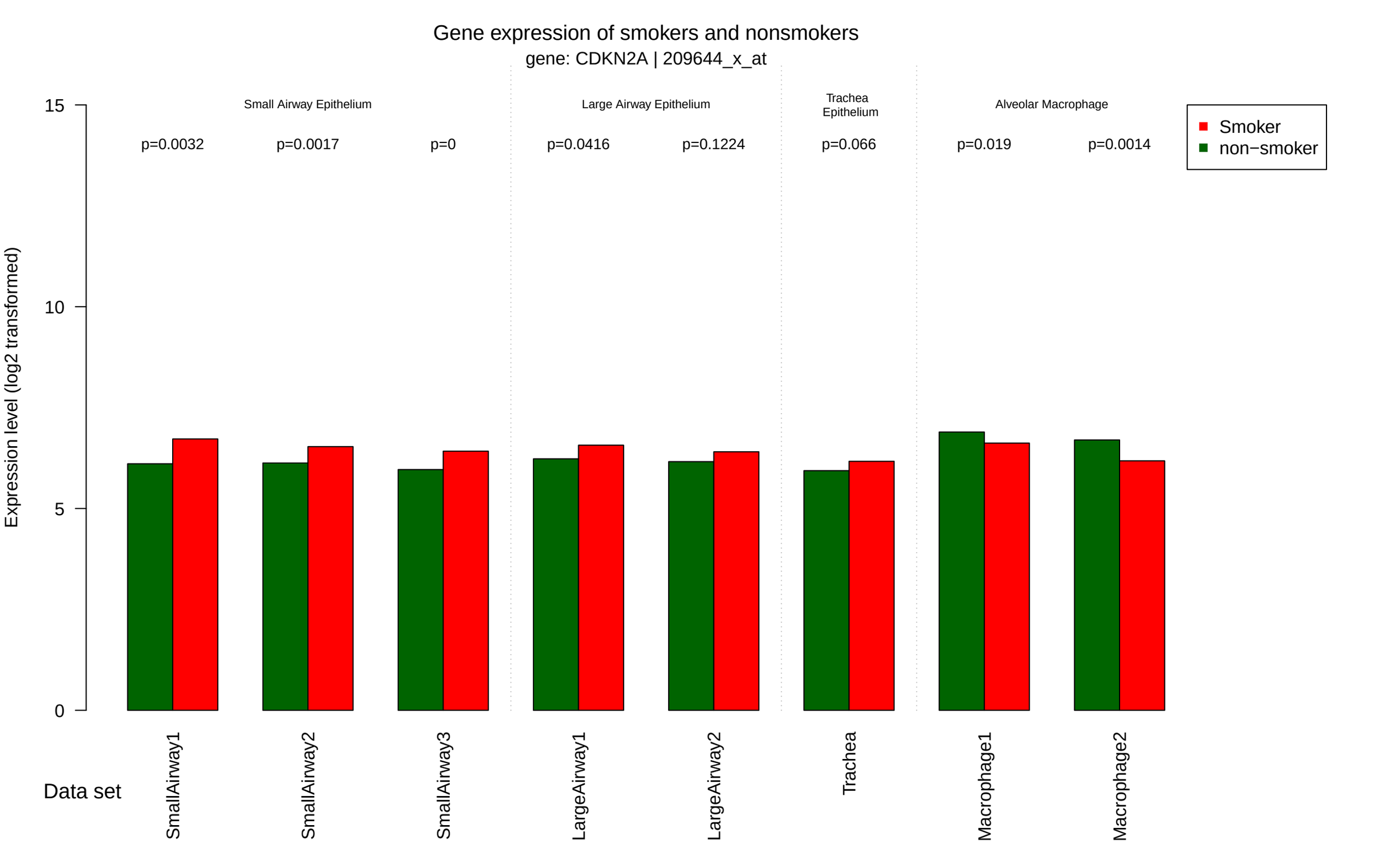
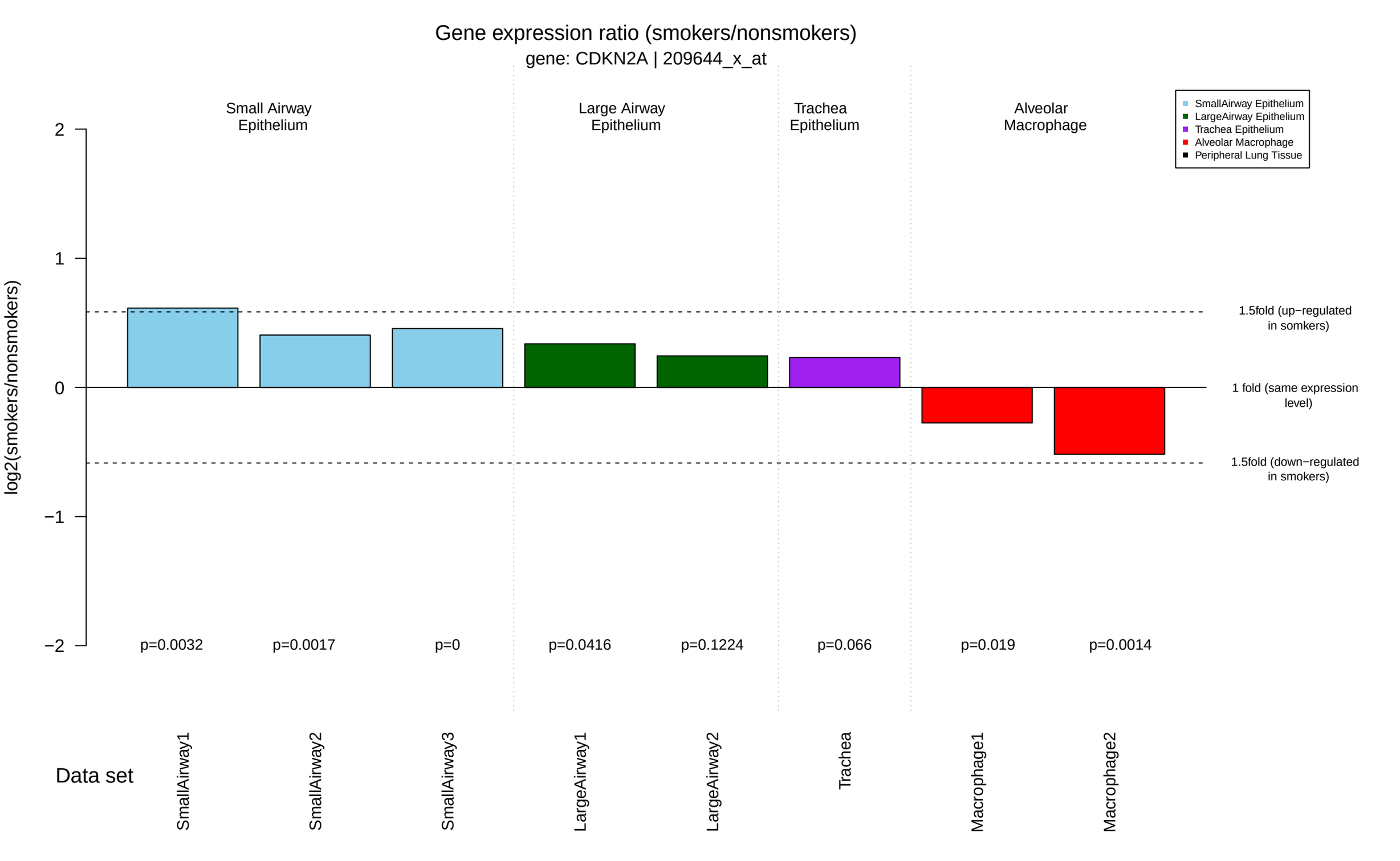
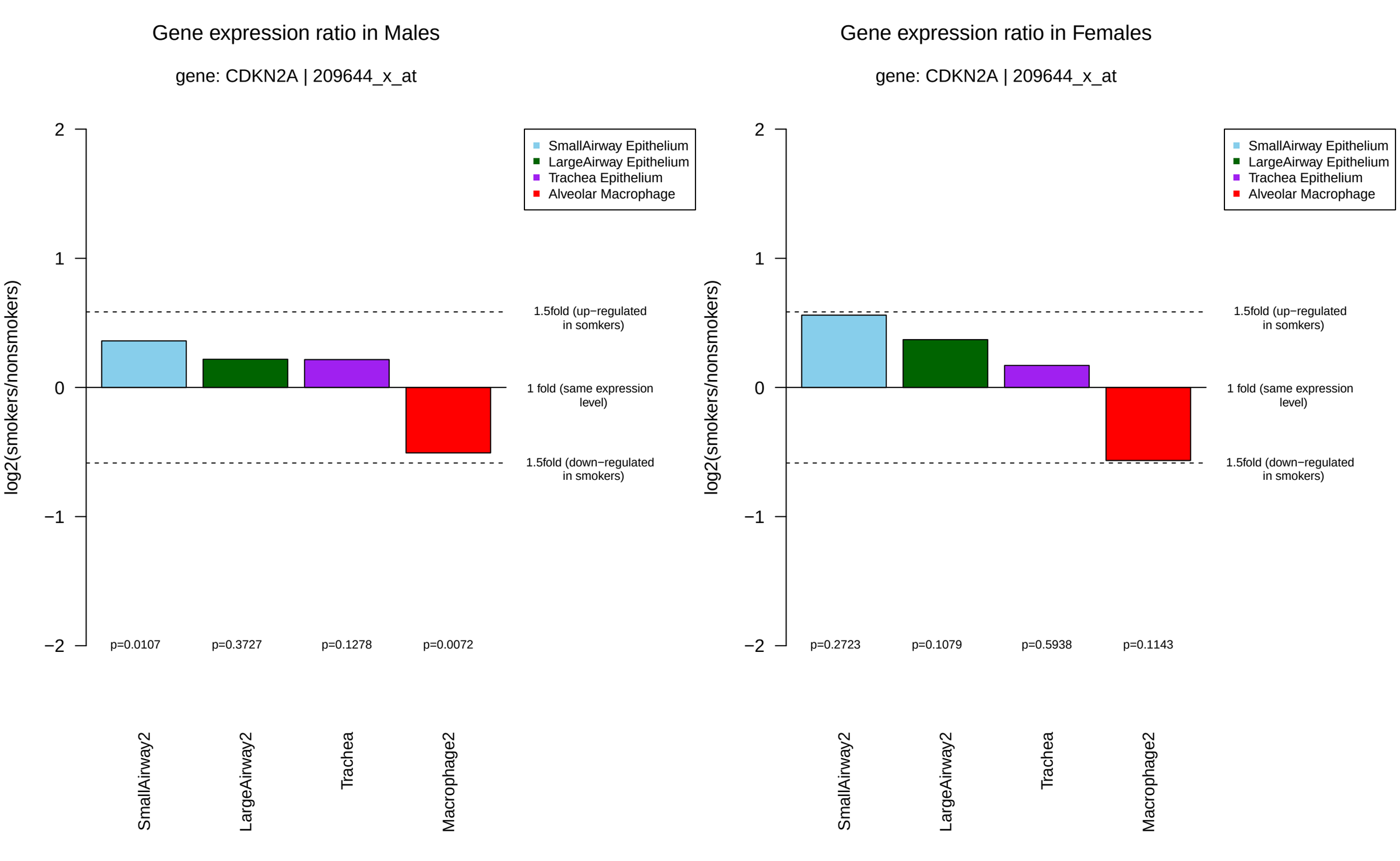
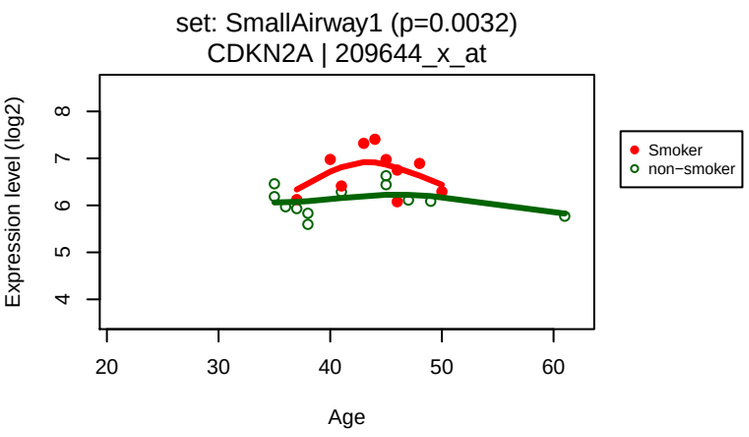
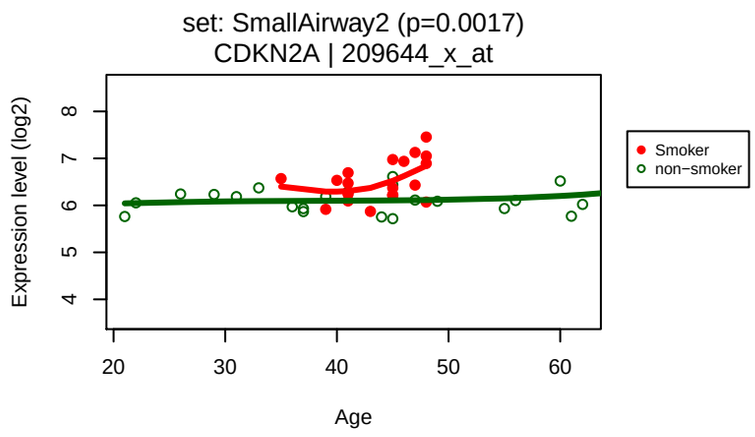
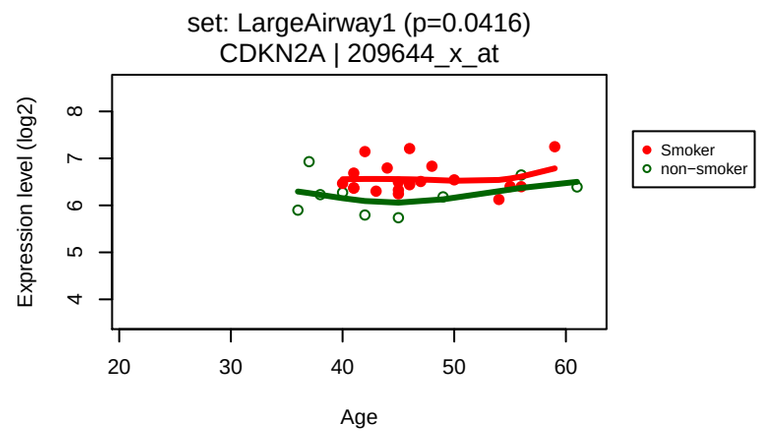

Gene expression of smokers and nonsmokers
gene: CDKN2A | 209644_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0032
p=0.0017
p=0
p=0.0416
p=0.1224
p=0.066
p=0.019
p=0.0014
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

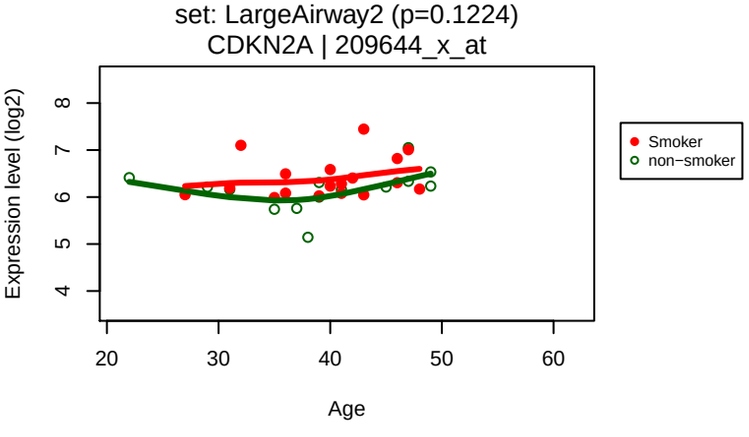

Gene expression ratio (smokers/nonsmokers)
gene: CDKN2A | 209644_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0032
p=0.0017
p=0
p=0.0416
p=0.1224
p=0.066
p=0.019
p=0.0014
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

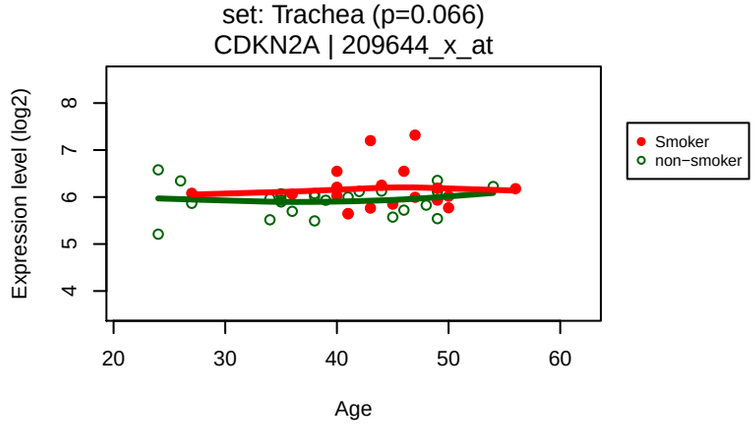

Gene expression ratio in Males
gene: CDKN2A | 209644_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0107
p=0.3727
p=0.1278
p=0.0072
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: CDKN2A | 209644_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2723
p=0.1079
p=0.5938
p=0.1143
SmallAirway2
LargeAirway2
Trachea
Macrophage2

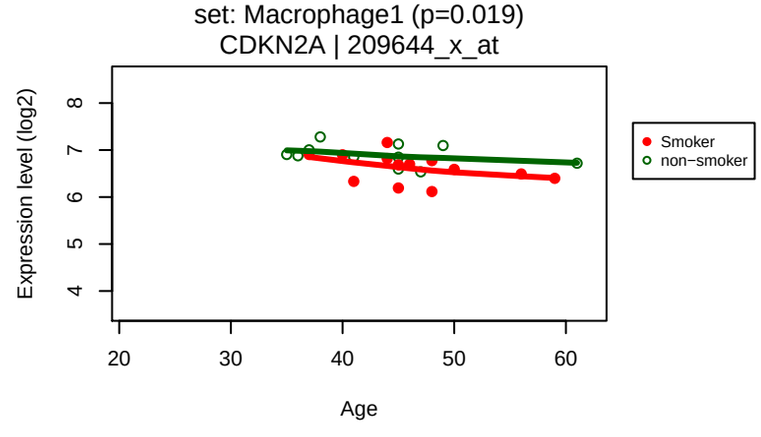

set: SmallAirway1 (p=0.0032)
CDKN2A | 209644_x_at
Expression level (log2)
Age
Smoker
non−smoker

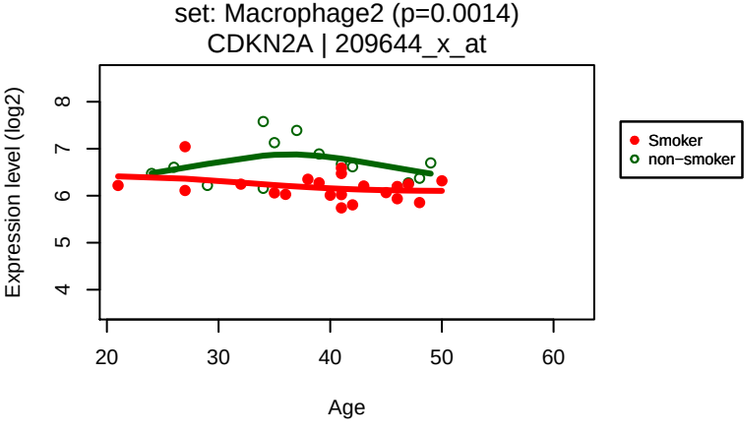

set: SmallAirway2 (p=0.0017)
CDKN2A | 209644_x_at
Expression level (log2)
Age
Smoker
non−smoker

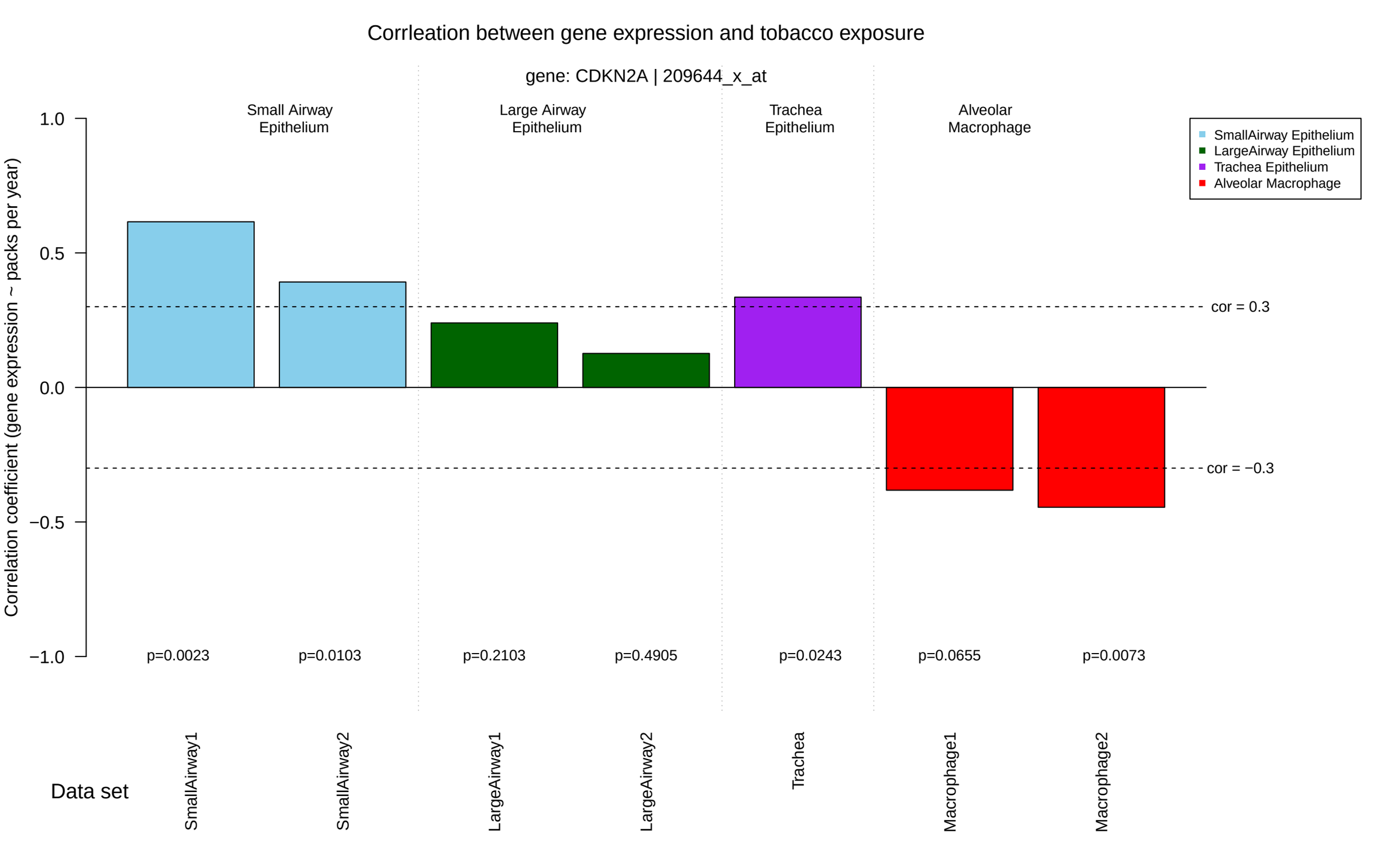

set: LargeAirway1 (p=0.0416)
CDKN2A | 209644_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1224)
CDKN2A | 209644_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.066)
CDKN2A | 209644_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.019)
CDKN2A | 209644_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0014)
CDKN2A | 209644_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: CDKN2A | 209644_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0023
p=0.0103
p=0.2103
p=0.4905
p=0.0243
p=0.0655
p=0.0073
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2